NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR

Odd Semester End-term Examination, 2025-26

Course Code: MAC01 | Full Marks: 60

Course Name: Mathematics-I | Time: 3 Hrs

- Answer any TEN questions.
- Symbols have their usual meanings.

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1. Test the convergence of the series

   (a) $2 - \frac{3}{2} + \frac{4}{3} - \frac{5}{4} + \cdots$, (b) $\sum_{n=1}^{\infty} \frac{x^n}{1+x^n}$, $(x > 0)$. [3+3] [CO#4] 1.2 1.1

2. (a) Show that the limit exists at the origin for the following function:

$$f(x,y) = \begin{cases} x\sin\left(\frac{1}{y}\right) + y\sin\left(\frac{1}{x}\right), & \text{if } xy \neq 0, \\ 0, & \text{if } xy = 0. \end{cases}$$

   (b) If $u = x\phi\left(\frac{y}{x}\right) + y\psi\left(\frac{y}{x}\right)$, prove that

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 0.$$

[3+3] [CO#1]

3. (a) Prove that

$$\left(\int_0^{\frac{\pi}{2}} \sqrt{\sin\theta}\, d\theta\right) \cdot \left(\int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{\sin\theta}}\right) = \pi.$$

   (b) Express

$$\int_0^1 t^m (1-t^n)^p \, dt$$

   in terms of Beta function. [4+2] [CO#2,3]

4. Using Taylor's theorem prove that

$$x - \frac{x^3}{3!} < \sin x < x - \frac{x^3}{3!} + \frac{x^5}{5!} \quad \text{and} \quad x - \frac{x^2}{2} < \log(1+x) < x, \quad \text{for } x > 0.$$

[3+3] [CO#1]

5. If $u = x + 2y + z$, $v = x - 2y + 3z$, and $w = 2xy - xz + 4yz - 2z^2$, then find the value of

$$\frac{\partial(u, v, w)}{\partial(x, y, z)}$$

Is there any relation between u, v, and w? If yes, then find the relation between them? [6] [CO#1]

6. Find the area bounded by the cardioide $r = a(1 - \cos\theta)$. [6] [CO#3]

7. By changing the order of integration, evaluate

$$\int_0^1 \int_x^1 \sin(y^2)\, dy\, dx.$$

[6] [CO#2]

8. Show that $\int_0^\infty e^{-x^2} dx$ converges and hence find its value. [4+2] [CO#2]

9. Find the volume under $z = x^2 + y^2$ over the region: $0 \le x \le 3, -1 \le y \le 1$. [6] [CO#3]

10. Evaluate

$$\iiint_W e^{(x^2+y^2+z^2)^{\frac{3}{2}}}\, dx\, dy\, dz$$

where $W = \{(x, y, z) : x^2 + y^2 + z^2 \le 1\}$. [6] [CO#3]

11. Find the area of that part of the cylinder $x^2 + y^2 = a^2$, which is cut off by the cylinder $x^2 + z^2 = a^2$. [6] [CO#2]

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NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR

Odd Semester End-term Examination, 2024-25

Course Code: MAC01 — Full Marks: 60

Course Name: Mathematics-I — Time: 3 Hrs

Date of Exam: 19.11.2024

- Answer any **TEN** questions. Symbols have their usual meanings.

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1. If $f(x) = (x-a)^m(x-b)^n$ where m and n are positive integers, show that 'c' in Rolle's theorem divides the segment $a \leq x \leq b$ in the ratio $m : n$. [6] [CO#1]

2. If $u = f(x^2+2yz, y^2+2xz)$ then prove that [6] [CO#2]
$$(y^2-zx)\frac{\partial u}{\partial x} + (x^2-yz)\frac{\partial u}{\partial y} + (z^2-xy)\frac{\partial u}{\partial z} = 0.$$

3. If $y_1 = 1-x_1, y_2 = x_1(1-x_2), y_3 = x_1x_2(1-x_3), \ldots, y_n = x_1x_2\ldots x_{n-1}(1-x_n)$, then find the Jacobian
$$\frac{\partial(y_1, y_2, \ldots, y_n)}{\partial(x_1, x_2, \ldots, x_n)}.$$
[6] [CO#2]

4. Show that the series $\dfrac{1}{(1+a)^p} - \dfrac{1}{(2+a)^p} + \dfrac{1}{(3+a)^p} - \cdots$, $a \geq 0$ is
   (i) absolutely convergent if $p > 1$.
   (ii) conditionally convergent if $0 < p \leq 1$. [3+3] [CO#3]

5. Test the convergence of the series by D'Alembert's ratio test
$$\left(\frac{1}{3}\right)^2 + \left(\frac{1.2}{3.5}\right)^2 + \left(\frac{1.2.3}{3.5.7}\right)^2 + \cdots$$
[6] [CO#3]

6. Find the length of the arc of the parabola $y^2 = 9x$, which is intercepted between the points of intersection of the parabola and the straight line $y = 3x$. [6] [CO#3]

7. Express
$$\int_0^1 t^m(1-t^n)^p dt,$$
where m, n, p are positive real numbers, in terms of Beta function and hence evaluate
$$\int_0^1 t^5(1-t^3)^9 dt.$$
[4+2] [CO#3]

8. Find the perimeter of the Cardioid $r = a(1 - \cos\theta)$. [6] [CO#3]

9. Evaluate

$$\iiint_E dV,$$

where E is the solid enclosed by the ellipsoid $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$. [6] [CO#3]

10. Evaluate

$$\iint_{D=\{(x,y)\in\mathbb{R}^2:0\leq x^2+y^2\leq 1\}} \sqrt{\frac{1-x^2-y^2}{1+x^2+y^2}}\, dx\, dy.$$

[6] [CO#3]

11. Evaluate [6] [CO#3]

$$\int_{y=0}^{1}\int_{x=y}^{1} \cos x^2 \ dx \ dy.$$

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NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR

Odd Semester End-term Examination, 2023-24

Course Code: MAC01	Full Marks: 60
Course Name: Mathematics-I	Time: 3 Hrs
Question Paper No.: NITDGP/	Date of Exam: 04.12.2023

- Answer all questions.
- Symbols have their usual meanings.

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1. (a) By Taylor's theorem prove that

$$1 + \frac{x}{2} - \frac{x^2}{8} < \sqrt{1+x} \quad \text{for } x > 0.$$

(b) Prove that between any two real roots of $e^x \sin x + 1 = 0$ there exist at least one real root of $\tan x + 1 = 0$. [3+3] [CO#1]

2. Show that $f_{xy}(0,0) \neq f_{yx}(0,0)$ for the following function [6] [CO#1]

$$f(x,y) = \begin{cases} xy & \text{for } |x| \geq |y| \\ -xy & \text{for } |x| < |y|. \end{cases}$$

3. Examine the following series for convergence/absolute convergence for all $x \in \mathbb{R}$

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots .$$

6 [CO#4]

4. Prove that

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{\sqrt{\pi}\,\Gamma(2n)}{2^{2n-1}\Gamma(n)}$$

for any natural number $n \in \mathbb{N}$. 6 [CO#2]

5. Examine the convergence of $\displaystyle\int_0^1 \frac{x^{p-1}}{1-x}\,dx$, $p \in \mathbb{R}$. 6 [CO#4]

6. Evaluate

$$\iiint_V \frac{dx\,dy\,dz}{(x^2+y^2+z^2)^{\frac{3}{2}}}$$

where V is the solid bounded by the spheres $x^2+y^2+z^2 \leq a^2$ and $x^2+y^2+z^2 \leq b^2$, where $a > b > 0$. 6 [CO#2]

7. Evaluate

$$\iint r^3 \, dr \, d\theta$$

over the region between $r = 2\sin\theta$ and $r = 4\sin\theta$. 6 [CO#2]

OR

By changing the order of integration, evaluate

$$\int_0^{2a} \int_{\frac{y^2}{4a}}^{3a-y} (x^2 + y^2) \, dxdy.$$

6 [CO#2]

8. Evaluate by Green's theorem

$$\oint_C \left[(\cos x \sin y - xy)dx + \sin x \cos y \; dy\right]$$

where C is the circle $x^2 + y^2 = 1$. 6 [CO#3]

9. Find the total work done of a moving particle in a force field given by

$$\vec{F} = (2x - y + z)\hat{\imath} + (x + y - z)\hat{\jmath} + (3x - 2y - 5z)\hat{k}$$

along a circle C in the xy plane $x^2 + y^2 = 9$. 6 [CO#3]

OR

Prove that $\vec{F} = \big(y^2 \cos x + z^3\big)\hat{\imath} + \big(2y \sin x - 4\big)\hat{\jmath} + \big(3xz^2 + 2\big)\hat{k}$ is a conservative force field. Find the scalar potential for $\vec{F}$. Also, find the work done in moving a particle in this field from $\big(0, 1, -1\big)$ to $\big(\frac{\pi}{2}, -1, 2\big)$. 6 [CO#3]

10. Evaluate

$$\iint_S \vec{F} \cdot \hat{n} \, dS,$$

where $\vec{F} = 18z\hat{\imath} - 12\hat{\jmath} + 3y\hat{k}$ and S is the part of the plane $2x + 3y + 6z = 12$ which is located in the first octant. 6 [CO#3]

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NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR

Odd Semester End-term Examination, 2022-23

Course Code: MAC01 | Full Marks: 60
Course Name: Mathematics-I | Time: 3 Hrs
Question Paper No.: NITDGP/ | Date of Exam: 20.02.2023

- Answer any 6 questions.
- Symbols have their usual meanings.

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1. (a) A function f is differentiable on $[0,2]$, and $f(0)=0, f(1)=2, f(2)=1$. Prove that $f'(c)=0$ for some $c \in (0,2)$. 5 [CO#1]

   (b) Verify Rolle's theorem for $f(x)=3+(x-1)^{1/3}$ in $0 \leq x \leq 2$. 5 [CO#1]

2. (a) Show that the following function is not continuous at $(0,0)$:

$$f(x,y)=\begin{cases}\dfrac{x^3+y^3}{x-y}, & x \neq y\\ 0, & x=y\end{cases}.$$

5 [CO#1]

   (b) If $u=\sin^{-1}\sqrt{\dfrac{x^{\frac{1}{3}}+y^{\frac{1}{3}}}{x^{\frac{1}{2}}+y^{\frac{1}{2}}}}$, then show that

$$x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x \partial y}+y^2\frac{\partial^2 u}{\partial y^2}=\frac{\tan u}{144}(13+\tan^2 u).$$

5 [CO#1]

3. (a) Prove that $\beta(m,n)=\frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$. Hence deduce that $\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$. 4+1 [CO#2]

   (b) Find the value of $\int_0^1 x^3(1-\sqrt{x})^5\,dx$. 5 [CO#2]

4. (a) Calculate the integral

$$\iint \frac{(x-y)^2}{x^2+y^2}\,dx\,dy$$

   over the circular region $x^2+y^2 \leq 1$. 5 [CO#2]

   (b) Using line integral, find the total work done in moving a particle in a force field given by

$$\vec{F}=3xy\hat{\imath}-5z\hat{\jmath}+10x\hat{k}$$

   along the curve $x=t^2+1, y=2t^2, z=t^3$ from $t=1$ to $t=2$. 5 [CO#3]

5. (a) Evaluate

$$\iint_S \vec{F} \cdot \hat{n} \, dS,$$

where $\vec{F} = yz\hat{i} + zx\hat{j} + xy\hat{k}$ and S is that part of the surface of the sphere $x^2 + y^2 + z^2 = 1$, which lies in the first octant. 5 [CO#3]

(b) Verify Green's theorem for

$$\oint_C (xy + y^2) \, dx + x^2 \, dy,$$

where C is the boundary of the closed region bounded by $y = x$ and $y = x^2$.

5 [CO#3]

6. (a) Evaluate

$$\iiint \frac{dx \, dy \, dz}{(x + y + z + 1)^3}$$

over a tetrahedron bounded by the coordinate planes and the plane $x+y+z = 1$.

5 [CO#2]

(b) Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, 1, 2)$. 5 [CO#3]

7. (a) Test the conditional and absolute convergence of the series

$$\sum_{n=1}^{\infty} \left(\frac{1 + nx}{n} \right)^n x^n,$$

where $x > 0$. 5 [CO#4]

(b) Test the convergence of the series $\sum_{n=1}^{\infty} u_n$ where

$$u_n = \sqrt{(n^4 + 1)} - \sqrt{(n^4 - 1)}.$$

5 [CO#4]

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Q. No. MAC 01 / 873 FIRST SEM : B.TECH / ODD — REG / (19 - 20)

2019-20

MATHEMATICS - I

MAC 01

Full Marks : 60 *Time* : 3 hours

The figures in the margin indicate full marks.

Symbols have their usual meanings

Answer any *six* questions. $10 \times 6 = 60$

1. (*a*) If $a < b$, prove that

$$\frac{b-a}{1+b^2} < (\tan^{-1}b - \tan^{-1}a) < \frac{b-a}{1+a^2}.$$

Hence deduce that $\frac{5\pi+4}{20} < \tan^{-1}2 < \frac{\pi+2}{4}$. 3 + 2

(*b*) Suppose ρ is the radius of curvature at a point of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and p is the length of the perpendicular from the centre to the tangent at that point. Then find a relation between ρ and p. 3 + 2

2. (*a*) Obtain the total length of the cardioid $r = a(1 + \cos\theta)$. Show that the arc of the upper half is bisected at $\theta = \frac{\pi}{3}$. 2 + 3

(*b*) Find the surface area of the solid generated by revolving the astroid

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$$

about the x–axis. 5

3. (*a*) Evaluate

$$\iint_R \sqrt{4x^2 - y^2}\, dx\, dy,$$

where $\mathcal{R}$ is the triangular region bounded by the straight lines $y = 0$, $x = 1$ and $y = x$. 5

(*b*) Change the order of integration in the integral

$$\int_0^1 dy \int_{\sqrt{y}}^{2-y} xy\, dx$$

and hence evaluate it. 3 + 2

4. (*a*) Evaluate

$$\iiint_W z^2 dx\, dy\, dz,$$

where W is the region bounded by the cylinder $x^2 + y^2 = a^2$, the paraboloid $z = x^2 + y^2$ and the xy-plane. 5

(*b*) Examine the convergence of the improper integral

$$\int_0^\infty \frac{dx}{x \log x}.$$ 5

5. (*a*) Show that the following function is not continuous at (0, 0)

$$f(x, y) = \begin{cases} \dfrac{x^3 + y^3}{x - y}, & x \neq y \\ 0, & x = y. \end{cases}$$ 4

(*b*) If $u = x + y + z$, $uv = y + z$, $uvw = z$ then find the value of $\frac{\partial(x,y,z)}{\partial(u,v,w)}$. 3

(*c*) Evaluate $\int_0^{\infty} \sqrt{x} e^{-\sqrt[3]{x}} dx$. 3

6. (*a*) If $u(x,y) = x\phi\left(\frac{y}{x}\right) + y\Psi\left(\frac{y}{x}\right)$, prove that

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0.$$ 5

(*b*) Test the absolute and conditional convergence of the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$ for every $x \in \mathbb{R}$. 5

7. (*a*) Find the constants a and b such that $\bar{f} = (axy + z^3)\hat{i} + \left(3x^2 - z\right)\hat{j} + (bxz^2 - y)\hat{k}$ is irrotational. Also, find the scalar potential of $\bar{f}$. 2 + 3

(*b*) Verify Green's theorem for $\int_C (xy + y^2)dx + x^2 dy$ where C is the closed curve made up of the line $y = x$ and the parabola $y = x^2$. 5